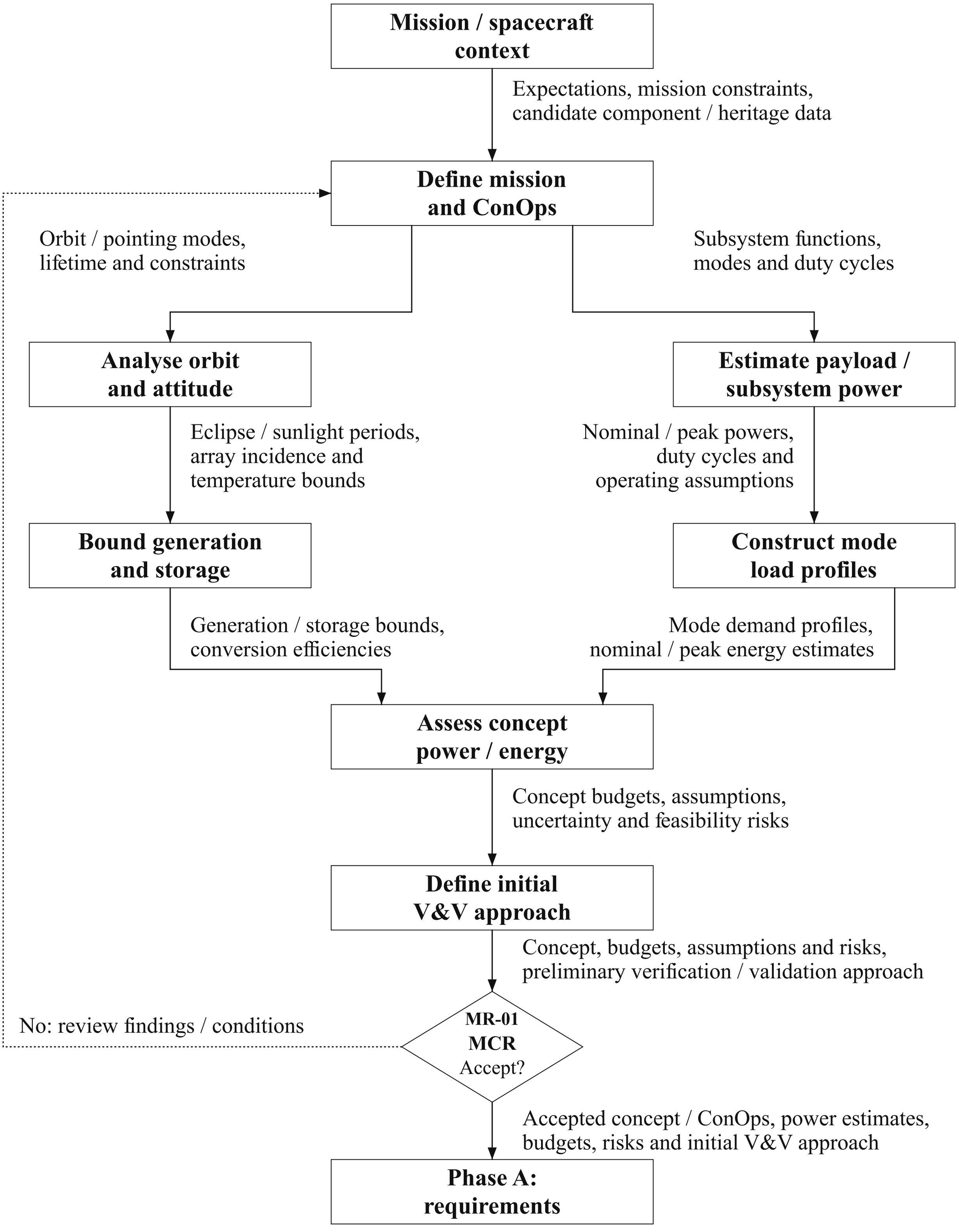
Pre-Phase A: mission concept and EPS feasibility



Phase A: EPS requirements and joint power feasibility

Accepted MCR basis and spacecraft inputs

Concept / ConOps, spacecraft allocations, sourced demand estimates and power bounds

Derive EPS requirements

Allocated EPS functions and electrical constraints

Stakeholder expectations, mission / spacecraft limits

Specify EPS interfaces

Define measurable performance criteria

Electrical interface limits, allocated subsystem demands and modes

Power / energy thresholds, units and operating conditions

Load schedules, peak demand, sunlight / eclipse energy

Model subsystem load scenarios

Assess joint power feasibility

Load assumptions and units, input data provenance

Feasibility findings, conflicts and margins

Validate requirements and resolve conflicts

Validated requirements and interfaces, trace links and conflict dispositions

Plan requirement verification

EPS requirements, interfaces, measurable criteria and joint feasibility findings

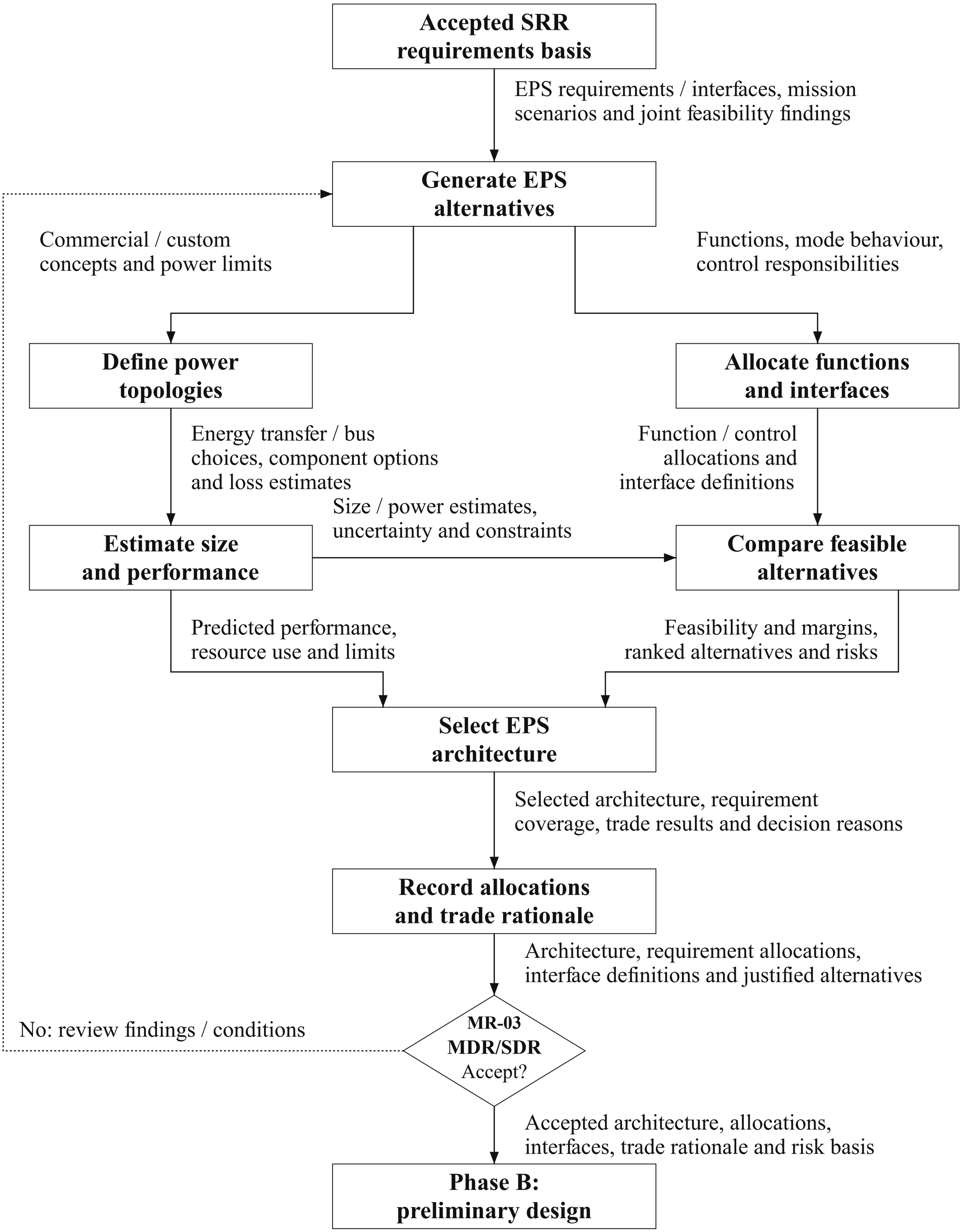
MR-02
SRR
Accept?

No: review findings / conditions

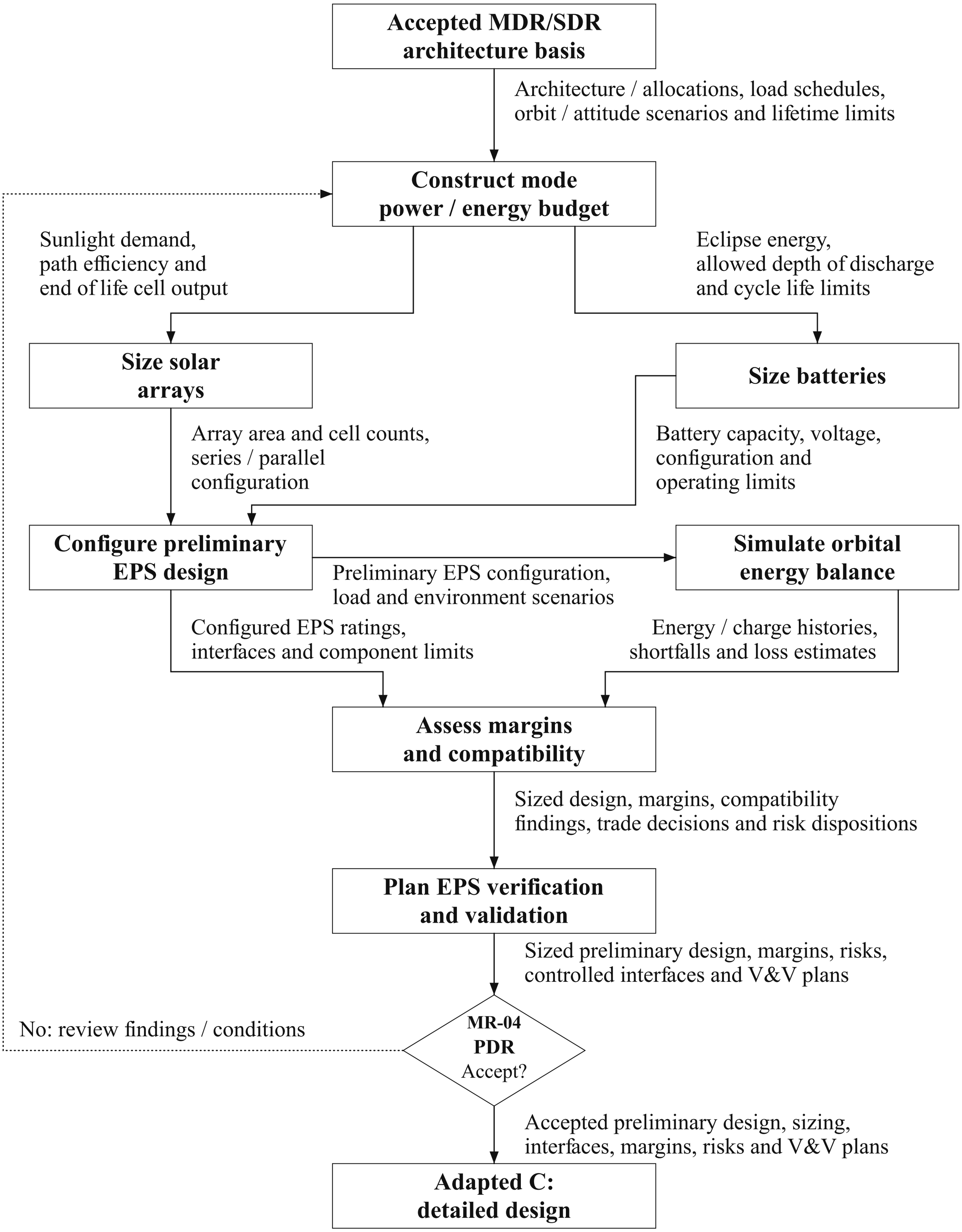
Accepted EPS requirements, interfaces, allocations and measurable verification criteria

Phase A: architecture

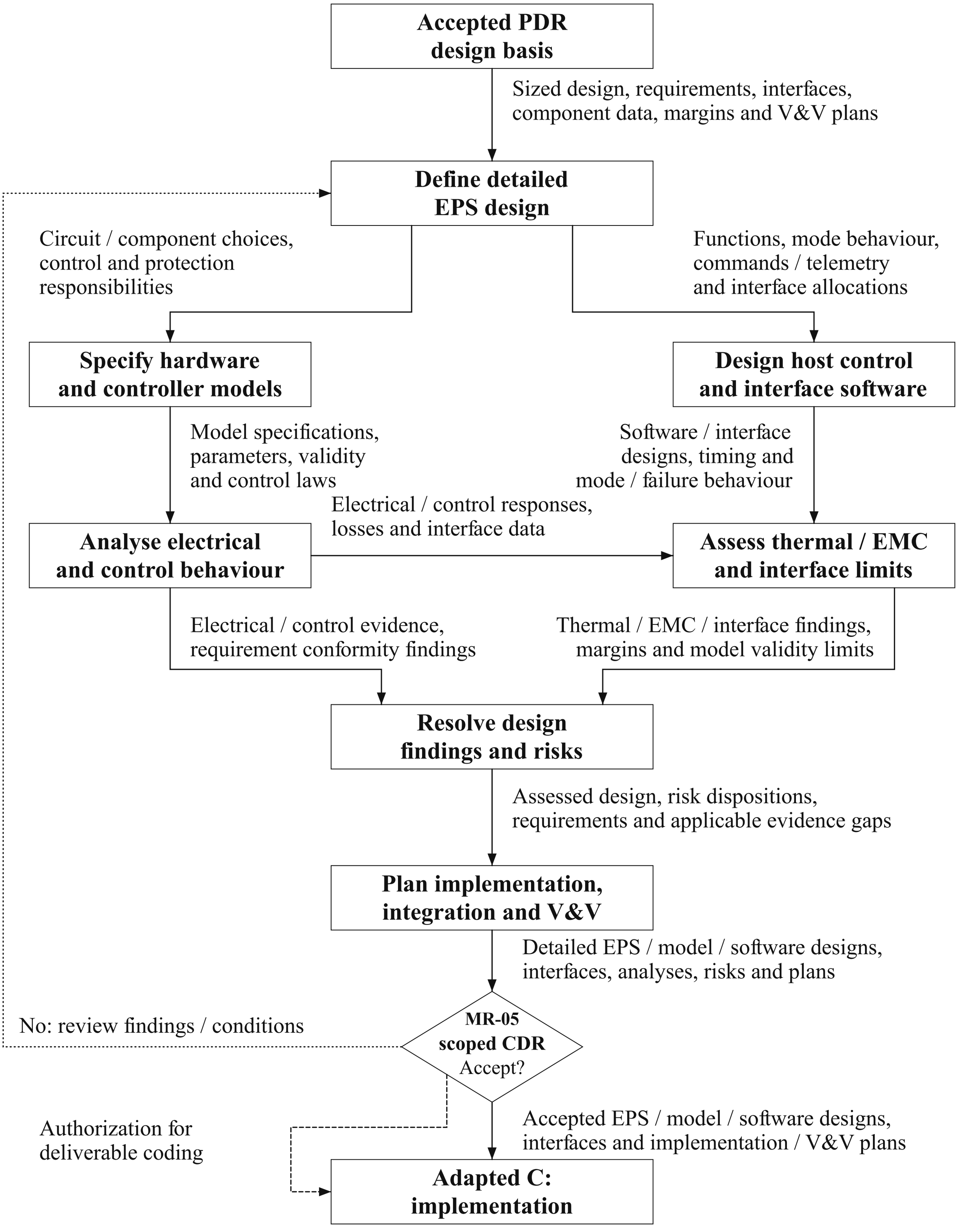
Phase A: EPS architecture and requirement allocation



Phase B: EPS sizing and preliminary design



Adapted C: detailed EPS design before deliverable coding



Adapted C: model and host software integration readiness

